MAHATMA GANDHI UNIVERSITY, KOTTAYAM

BSc (Honours) Computer Science
FIFTH SEMESTER EXAMINATION
(2024 Admission Onwards)

MG5DSECSC301: RESOURCE OPTIMIZATION TECHNIQUES

MODEL QUESTION PAPER

Duration: 2 Hrs Maximum Marks: 70

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A: Very Short Answer Questions.
(Answer **All** Questions. Each question carries **2** marks.)

1. What are the exceptional cases that may arise while solving an LPP graphically? [U] [CO1]
2. A primal LPP has 3 decision variables and 5 constraints. Apply the properties of duality to determine the number of variables and constraints in its dual problem. [A]CO2]
3. What is a prohibited assignment problem? [K] [CO3]
4. Distinguish between PERT and CPM. [An] [CO4]
5. List any two steps involved in the Operations Research methodology. [K] [CO1]

[5 × 2 = 10]

PART B: Short Answer Questions
(Answer **6** out of **8** questions, each question carries **5** marks.)

6. Explain the nature, characteristics and methodology of Operations Research. [U] [CO1]
7. Estimate the following LPP graphically:
   Maximize
   $Z = 5x_1 + 4x_2$
   Subject to
   $6x1 + 4x2 \leq 24$
   $x1 + 2x2 \leq 6$
   $x1, x2 \geq 0$ [U] [CO1]
8. Distinguish slack and surplus variable. [U] [CO1]
9. Formulate the dual problem of the following primal LPP:
   Maximize
   $Z=3x1+5x2$
   Subject to
   $x1 + 2x2 \leq 8$
   $3x1 + 2x2 \leq 12$
   $x1, x2 \geq 0$ [A] [CO2]
10. Obtain an initial basic feasible solution for a given transportation problem using the Vogel's Approximation Method. Explain the steps involved in the method. [A] [CO3]
11. Illustrate the step-by-step procedure for applying the North-West Corner Method to obtain an initial basic feasible solution in a transportation problem. [A] [CO3]
12. A set of jobs has to be processed through two machines in the order M1–M2. Given the processing times for each job, determine the optimal sequence and minimum total elapsed time using Johnson's rule. [A] [CO4]
13. Illustrate the concept of Maintenance Crew Scheduling and analyse its main objectives. [An] [CO4]

[6 × 5 = 30]

PART C : Essay Questions

(Answer **2** out of **3** questions, each question carries **15** marks.)

14. Solve the following LPP using the simplex method and determine the optimal solution:

Maximize

$Z = 3x_1 + 5x_2$

Subject to

$x_1 + 2x_2 \leq 8$

$3x_1 + 2x_2 \leq 12$

$x_1, x_2 \geq 0$

Show the complete simplex table at each iteration. [A] [CO2]

15. Solve the following assignment problem using the Hungarian method and determine the minimum-cost assignment:

	J1	J2	J3	J4
W1	9	2	7	8
W2	6	4	3	7
W3	5	8	1	8
W4	7	6	9	4

[A] [CO3]

16. For the project activities and their precedence relationships given below, construct the network, determine the critical path and calculate the project duration using CPM. Also determine the slack of the non-critical activities.

Activity	**Predecessor**	**Duration**
A	–	4
B	–	6
C	A	5
D	A	3
E	B, C	4
F	C	6
G	D, E	5
H	F, G	3

[An] [CO4]

[2 ×15 = 30]

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU – UGP (HONOURS) FIFTH SEMESTER EXAMINATION
MG5DSECSC302 - ARTIFICIAL INTELLIGENCE
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Max. Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions
Answer **all** questions
Each question carries **2** marks

1. Define Artificial Intelligence. [K] [CO1]
2. List two applications of Artificial Intelligence. [K] [CO1]
3. State the search order followed by Depth-First Search. [U] [CO2]
4. Differentiate between Prior Probability and Posterior Probability. [U] [CO4]
5. A person has a membership value of 0.8 in the fuzzy set Young. Interpret this value. [U] [CO4]

Part B

Short Answer Questions
Answer any **6** questions
Each question carries **5** marks

6. Define control strategy in AI problem solving. [U] [CO1]
7. Demonstrate Breadth-First Search for a given tree and trace the order of node expansion. [U] [CO2]
8. Analyze the advantages and limitations of Breadth-First Search. [A] [CO2]

9. Demonstrate the complete search path of A* Search for a given search tree using f(n)=g(n)+h(n). [A] [CO2]
10. Compare Hill Climbing and Greedy Best-First Search with respect to node selection, memory usage, and search behavior. [A] [CO2]
11. Represent the following statements using First-Order Predicate Logic: All humans are mortal. Socrates is a human. Some students are intelligent. [U] [CO3]
12. Explain the working principle of Backward Chaining with a suitable example. [U] [CO3]
13. A fuzzy controller receives a crisp value of 70 as its input. Explain how fuzzification converts this numerical input into fuzzy information. Analyze the significance of membership values obtained during this process. [An] [CO4]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

14. Discuss how Expert Systems can be used as a knowledge-based tool in Environmental Management. Explain the process of using expert knowledge for environmental problem solving with suitable examples. [U] [CO1]
15. Analyze the role of universal and existential quantifiers in First-Order Predicate Logic and illustrate their use in representing real-world knowledge. [An] [CO3]
16. Analyze the complete process of fuzzification and defuzzification in a fuzzy system. Use a suitable example to support your analysis. [An] [CO4]